

Early Detection of Diabetes Risk Among Indian Women



Mitch Daniels School of Business

ABSTRACT

This project uses clinical and demographic data from Pima Indian women over 21 to explore how data analytics can support early detection of diabetes. Framed as a **classification** problem, we identified strong connections between glucose, BMI, insulin, and age, with glucose being the most predictive feature. Using **logistic regression** as our baseline model and more advanced models with **K-fold cross-validation**, we evaluated **sensitivity** to build reliable predictors. Our findings highlight how data-driven tools can inform early intervention and improve public health outcomes, especially in underserved populations.

BUSINESS PROBLEM

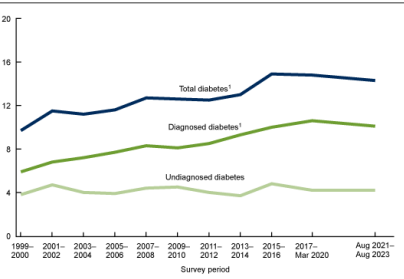
Diabetes affects over **588 million** people worldwide and many remain undiagnosed. Early prediction is critical to prevent complications such as kidney failure, heart disease, and vision loss, while reducing long-term healthcare costs.

However, late diagnosis is common, especially in underserved populations. Many healthcare systems rely on **reactive testing** rather than **proactive screening**, leading to higher costs and worse outcomes.

Our project uses **machine learning** to predict diabetes risk before onset using clinical indicators like glucose, BMI, insulin, blood pressure, and age. This enables earlier intervention and better resource allocation.

Business and Policy Relevance

- Healthcare:** Data-driven screening supports early detection and personalized care
- Insurance:** Improves preventive coverage and cost efficiency
- Policy:** Guides targeted programs and improves access for high-risk populations



[1] Trends in age-adjusted prevalence of total, diagnosed, and undiagnosed diabetes in adults age 20 and older: United States, 1999–2000 through August 2021–August 2023

RESEARCH QUESTION

Can we accurately predict the likelihood of diabetes in Pima Indian women over the age of 21 using clinical and demographic indicators?

ANALYTICS PROBLEM FRAMING

Problem:

Predict whether an individual is **at high risk for diabetes** using clinical and demographic data

Assumptions:

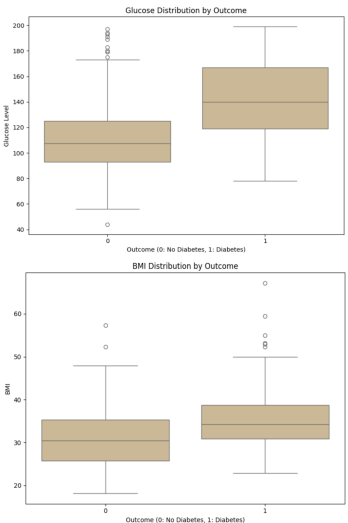
- The dataset accurately represents relevant risk factors
- Variables such as **glucose**, **BMI**, and **insulin** are reliable predictors of diabetes likelihood

Success Metrics:

- Sensitivity:** Correctly identifying **true positive (diabetic)** cases. Prioritizing it helps **catch high-risk patients early**, reducing missed diagnoses and improving health outcomes

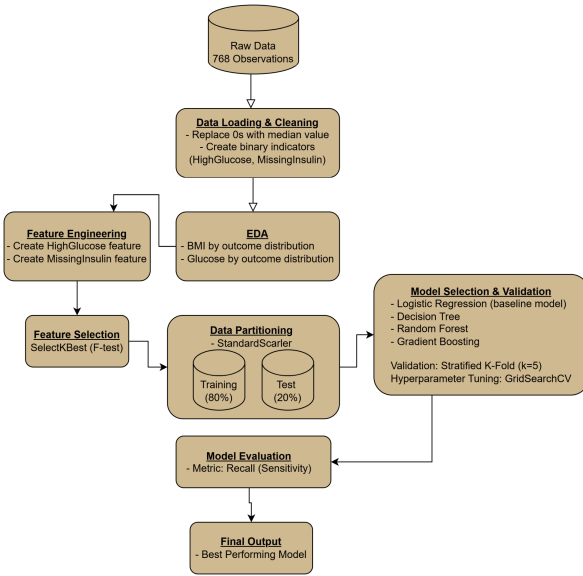
DATA & EDA

Dataset: Kaggle – Diabetes Dataset
Source: National Institute of Diabetes and Digestive and Kidney Diseases
Population: Females, age 21+, Pima Indian heritage
Response Variable: Outcome
Variables(11): Glucose, BMI, Age, Pregnancies, Insulin, BloodPressure, SkinThickness, DiabetesPedigreeFunction, AgeGroup, HighGlucose, MissingInsulin



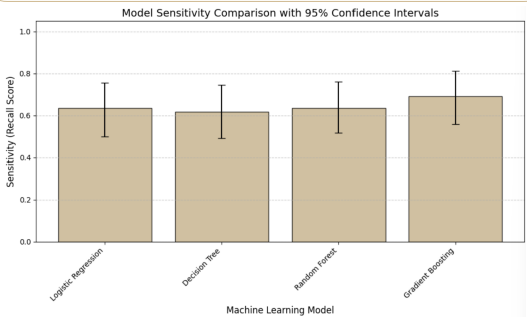
Connection to Problem: These relationships validate that measurable clinical indicators can predict diabetes risk. This supports the project's goal of building a machine learning model for early detection and proactive healthcare intervention

METHODOLOGY



MODELS PERFORMANCE AND EVALUATION

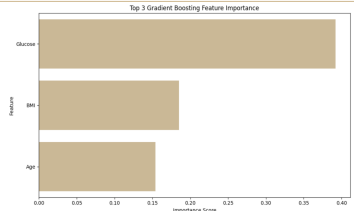
- Model performance was evaluated using **Sensitivity (Recall)**, focusing on the proportion of true positive diabetes cases correctly identified by the model
- Gradient Boosting** achieved the highest sensitivity with the lowest variance, making it the most consistent model — though the improvement was not statistically significant
- Confidence intervals** reinforce its reliability across test samples
- The model correctly identified approximately **7 out of 10** diabetes cases, reaching a **sensitivity of 0.69** despite class imbalance



- Useful for healthcare providers to **flag high-risk individuals**
- Future work: should **expand dataset size** and other **demographic data**

BUSINESS IMPLICATIONS

Gradient Boosting ranked **Glucose, BMI, and Age** as the top predictors. These predictors support smarter screening and strategic decisions across patient care, healthcare operations, and public health policy



Patient Impact

- 75% accuracy** enables early alerts for proactive changes
- Identifies **7 in 10** high-risk patients, limiting complications

Business and Healthcare

- Targets resources to high-risk patients
- Reduces costs and improves ROI for prevention

Policy and Public Health

- Guides funding and education in high-risk areas
- Promotes equity by identifying underserved populations

CONCLUSIONS & FUTURE WORK

Conclusions

- Our Gradient Boosting model achieved strong performance, correctly identifying **7 in 10** diabetes cases with high sensitivity
- Glucose, BMI, and Age emerged as the top predictors, reinforcing their clinical relevance and guiding targeted screening strategies

Future Work

- These insights can support healthcare professionals in identifying high-risk individuals earlier, enabling more proactive testing, personalized care, and efficient resource allocation
- Future work will focus on expanding the dataset, refining model parameters, and validating performance in real-world clinical settings to support integration into decision-making tools

ACKNOWLEDGEMENTS

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REFERENCES

[1] Total diagnosed and undiagnosed diabetes: National Center for Health Statistics, National Health and Nutrition Examination Surveys, 1999–2000 through August 2021–August 2023.